The Rules of Exponents

Exponents are used to show repeated multiplication of a number called the base.

For example, 2^5 means $2 \times 2 \times 2 \times 2 \times 2$. The number 2 is the base, and the exponent is 5. There are five 2s in the multiplication problem. To find the value, complete the multiplication: $2 \times 2 \times 2 \times 2 \times 2 = 32$

Variables can also be raised to a power. n^5 means $n \times n \times n \times n \times n$.

You can use the rules of exponents to simplify expressions with exponents. The rules are written with variables to show that they apply in all situations. They work whether the base is a number or a variable.

Study the rules and the examples that follow. Knowing how to simplify expressions with variables can make your work easier and save you valuable time on tests.

Rule	Meaning
$x^a \cdot x^b = x^{a+b}$	When multiplying the same bases, keep the same base and add the exponents.
$\frac{x^a}{x^b} = x^{a-b}$	When dividing the same bases, keep the same base and subtract the exponents.

Example 1

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Simplify $y^3 \times y^2$.
The bases are the same. Keep the same base and add the exponents.
$y^3 \times y^2 = y^{3+2} = y^5$

Example 2

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Simplify $n^6 \div n^4$.
The bases are the same. Keep the same base and subtract the exponents.
$n^6 \div n^4 = n^{6-4} = n^2$

The second rule can cause two special situations to occur.

$\frac{4^3}{4^2} = 4^{3-2} = 4^1$ What does a number raised to the first power equal? The exponent tells you to write the base only one time. A number raised to the first power equals itself. $4^1 = 4$

$\frac{2^3}{2^3} = 2^{3-3} = 2^0$ What does a number raised to the zero power equal? Any number divided by itself equals 1. $2^0 = 1$

$X^0 = 1$

When following the rules of exponents, you may get an answer with a negative exponent. At first glance, a negative exponent doesn't make sense. How can you raise a number to a negative power? In this lesson, you will learn how to interpret a negative exponent.

The first example is solved in two ways. As you can see, the rules of exponents are connected to canceling.

Example 3

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Using the rules of exponents: $\frac{a^5}{a^3} = a^{5-3} = a^2$

Using canceling: $\frac{\cancel{a}\cdot\cancel{a}\cdot\cancel{a}\cdot a\cdot a}{\cancel{a}\cdot\cancel{a}\cdot\cancel{a}} = a^2$

What happens when the exponent in the denominator is greater than the exponent in the numerator? The exponent is negative. Use canceling to find its meaning.

Example 4

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Using the rules of exponents: $\frac{a^3}{a^5} = a^{3-5} = a^{-2}$

Using canceling: $\frac{\cancel{a}\cdot\cancel{a}\cdot\cancel{a}}{\cancel{a}\cdot\cancel{a}\cdot\cancel{a}\cdot a\cdot a} = \frac{1}{a^2}$

From this example, you can see that $a^{-2} = \frac{1}{a^2}$.

$X^{-b} = \frac{1}{X^b}$

Example 4 reveals a key idea. You can change a negative exponent to a positive exponent by moving the base number and its exponent to the other side of the fraction bar.

Example 5

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x^{-3} becomes $\frac{1}{x^3}$.

$\frac{1}{2^{-4}}$ becomes $\frac{2^4}{1}$, which equals 16.

Tip

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Rule	Meaning
$(x^a)^b = x^{a \times b}$	When a base with an exponent is raised to a power, keep the same base and multiply the exponents. $(3^2)^3 = 3^6 = 729$
$(xy)^a = x^a y^a$ $\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$	When a product or a quotient is raised to a power, you can distribute the exponent. $(3x)^2 = 9x^2$ $\left(\frac{2}{5}\right)^3 = \frac{8}{125}$

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Lesson 1: Exponents and Roots

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